

Worst-case vulnerability assessment and mitigation model of urban utility tunnels

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ARTICLE INFO

Keywords:

Utility tunnels
Interdependencies
Urban infrastructure systems
Worst-case vulnerability

ABSTRACT

Constructing utility tunnels would bring new threats, among which the terrorist threat has been identified as the most serious one. Because utility tunnels increase geographical interdependencies among urban infrastructure systems (UISSs), and attacking a tunnel segment could cause simultaneous failure of all its carried systems. This paper models the terrorist threat as worst-case failure, and formulates a tri-level mathematical model to assess and mitigate the worst-case vulnerability of urban utility tunnels that carry interdependent UISSs. Exact solution of the model is obtained by a column-and-constraint generation based decomposition algorithm. Real utility tunnels that carry interdependent power and water systems in Tianjin Eco-city, China, are used to demonstrate the proposed approach. Results show that there exists a threshold value of the defense budget B_D . If B_D is less than that threshold, the utility tunnels greatly benefit the attacker and increase the power and water systems' worst-case vulnerability. If B_D is equal to that threshold, the utility tunnels almost do not affect the worst-case vulnerability. If B_D is larger than that threshold, the utility tunnels benefit the defender and reduce the worst-case vulnerability.

1. Introduction

Urban infrastructure Systems (UISSs), such as power, water and transportation systems, are not isolated but physically, geographically, cyber and logically interdependent [12,30,39]. Some interdependencies are positive in normal operation [30]. For example, the physical dependency of a water pump station on an electric load substation facilitates the water delivery; the cyber interdependency between an electric power system and its SCADA (supervisory control and data acquisition) system makes the power flow controlled better. However, under disruptions, as one system failure can propagate to its interdependent systems, some interdependencies aggravate the system-level vulnerability. In the literature, the vulnerability definition and metric are not unique [5,20]. This article interprets that the vulnerability is associated with a specific event and is quantified as system functionality drop incurred by the event [32].

Two systems are geographically interdependent if “a local environmental event can create state change in both systems” [39]. According to this definition, geographical interdependency emerges under a local environmental event or a disruptive event. Hence, modeling

geographical interdependency should be event-specific. In the literature, many researchers have studied this type of interdependency under natural hazards, which could affect large-scale area and cause simultaneous failure of components in different systems, such as vulnerability analysis of interdependent systems under earthquakes in Shelby County, USA [13,18], and interdependent power and gas transmission systems in Europe [38], and hurricane vulnerability for power and gas transmission systems in Harris County, USA [31]. Note that components in these transmission systems are usually not very close to each other, but due to the large-scale and extensive impact of natural hazards, different systems' components, even far from each other (several to hundreds of kilometers), could still be damaged simultaneously so that the geographical interdependency emerges. Readers may question this understanding of geographical interdependency, and think that the simultaneous failure of different components far from each other is not directly ascribable to geographical interdependency, but instead on the fragilities of the considered components to the given hazard intensity measure, and geographical interdependency is associated to a concept of proximity, when components of different system fail together because they share the same

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<https://doi.org/10.1016/j.ress.2020.106856>

Received 26 February 2019; Received in revised form 8 July 2019; Accepted 26 January 2020

Available online 18 February 2020

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Model parameters

$k \in \kappa$	set of all UIS networks
$n \in N^k$	set of nodes in UIS network k
$e \in E^k$	set of edges in UIS network k
$n \in N^T$	set of tunnel nodes in the network of the utility tunnels
$e \in E^T$	set of tunnel segments in the network of the utility tunnels
$n \in N^{S,k}$	set of source nodes in UIS network k
$n \in N^{D,k}$	set of demand nodes in UIS network k
$e \in E_n^{I,k}$	set of edges in E^k oriented to node n in UIS network k
$e \in E_n^{O,k}$	set of edges in E^k oriented from node n in UIS network k
$\tilde{N}^k \subseteq N^k$	set of the damageable nodes in UIS network k
$\tilde{E}^k \subseteq E^k$	set of the damageable edges in UIS network k
$M^k \cap \tilde{E}^k$	set of the damageable edges in UIS network k that are not fully carried by utility tunnels
$\tilde{N}^T \subseteq N^T$	set of the damageable tunnel nodes in the network of the utility tunnels
$\tilde{E}^T \subseteq E^T$	set of the damageable tunnel segments in the network of the utility tunnels
$F_{i,j}^{k \rightarrow l}$	set of ordered pairs (i, j) associated with node i in UIS network k and component j in UIS network l , and component j will fail if node i cannot provide sufficient supply
$o^k(e), d^k(e)$	origin and destination nodes of edge e in UIS network k
f_e^k	capacity for edge e in UIS network k
s_n^k	maximum supply for node n in UIS network k
d_n^k	required demand for node n in UIS network k
$I_{e,n}^k \in I^k$	1 if edge e in UIS network k passes through tunnel node n , and 0 otherwise
$J_{e,e'}^k \in J^k$	1 if edge e in UIS network k is partially carried inside tunnel segment e' , and 0 otherwise
$M_e^k \in M^k$	1 if edge e in UIS network k is not fully carried by utility tunnels, and 0 otherwise
η^k	vulnerability weighting factor for UIS network k
$c_n^{D,k}$	cost for protecting the damageable node n in UIS network k
$c_e^{D,k,M}$	cost for protecting the damageable and not-fully-carried

$c_n^{D,T}$	edge e in UIS network k
$c_e^{D,T}$	cost for protecting the tunnel node n in the network of the utility tunnels
B_D	cost for protecting the tunnel segment e in the network of the utility tunnels
B_D	total defense budget
$c_n^{A,k}$	cost of attacking node n in UIS network k
$c_e^{A,k,M}$	cost for attacking not-fully-carried edge e in UIS network k
$c_n^{A,T}$	cost for attacking tunnel node n in the network of the utility tunnels
$c_e^{A,T}$	cost for attacking tunnel segment e in the network of the utility tunnels
B_A	total attack budget

Decision variables

w_n^k	1 if the defender protects the damageable node n in UIS network k , and 0 otherwise
$w_e^{k,M}$	1 if the defender protects the damageable and not-fully-carried edge e in UIS network k , and 0 otherwise
w_n^T	1 if the defender protects the tunnel node n in the network of the utility tunnels, and 0 otherwise
w_e^T	1 if the defender protects the tunnel segment e in the network of the utility tunnels, and 0 otherwise
x_n^k	0 if damageable node n in UIS network k is attacked, and 1 otherwise
$x_e^{k,M}$	0 if damageable edge e that is not fully carried by utility tunnels in UIS network k is attacked, and 1 otherwise
x_n^T	0 if damageable tunnel node n is attacked in the network of the utility tunnels, and 1 otherwise
x_e^T	0 if damageable tunnel segment e is attacked in the network of the utility tunnels, and 1 otherwise
s_n^k	real node supply for node n in UIS network k
d_n^k	real node demand for node n in UIS network k
f_e^k	real edge flow for edge e in UIS network k
$\delta_{i,j}^{k \rightarrow l}$	1 if there is a physical interdependency from node i in UIS network k to component j in UIS network l

geographic location. But this view raises the following question, based on the 911 event stated in the next paragraph and several other similar events introduced by Ouyang [27], how close for components in proximity can be called “geographically interdependent”? It is difficult to provide a clear answer. Instead of addressing this question, this paper simply interprets “geographical interdependency” as event-specific based on its original definition by Rinaldi et al. [39].

Besides natural hazards, some other events, such as terrorist or intentional attacks, can also lead to simultaneous failure of different systems' components in the same region. For example, the New York 911 terrorist attack resulted in the full damage of the two buildings WTC1 and WTC2, with their debris damaging many infrastructure components (such as water pipes, power substations and telecommunication facilities, and subway tunnels) within 0.21 km from the attack center. This type of event is usually modelled as localized attack [25,29,34], where the components in different systems that are located within the same localized attack area could fail simultaneously. This type of attack has been extensively investigated in the literature to address geographical interdependencies for interdependent systems [24,29,34]. Note that different from natural hazards, the geographical interdependency in this type of events emerges when different systems' components are co-located within an attack influence area, which usually has a radius of several meters to several kilometers.

The above studies capture the geographical interdependencies mainly by considering some particular “environmental events”, so that different systems' components are co-located within the event influence

area. In practice, some components in different systems could be very close to each other, less than several meters or just inside the same small space. In this case, normal attack means or techniques, which are usually modelled to cause only one component damage [3], could result in simultaneous failure of components in different systems. In fact, the extreme spatial proximity of different UISs' components has been intentionally designed in practice through constructing utility tunnels in many modern cities. A utility tunnel is an underground structure containing one or more utilities or infrastructure systems, permitting installation, maintenance and removal of the systems without the necessity of cutting streets [9]. Apparently, constructing utility tunnels has many advantages [7,10,23,43], such as avoiding continual cutting of pavements, facilitating the installation, inspection, replacement and maintenance, exploiting limited urban underground space more efficiently, and improving urban appearance. Because of those advantages, utility tunnels have been widely constructed in many modern cities [22]. In China, new utility tunnels with a total length of 1000 km were constructed in 2015, and the construction length became more than 2000 km in 2016. However, constructing utility tunnels can bring new threats. By conducting a questionnaire survey, Canto-Perello [9] identified six types of threats on utility tunnels, including disgruntled employee, common criminals, terrorist, violent activities, vandals and urban explorers, and found the terrorist threat the most serious. This is because utility tunnels increase geographical interdependencies among UISs, so that attacking a tunnel segment could cause simultaneous failure of all its carried systems.

To complement the above survey-based qualitative work, this paper will make a quantitative analysis for the terrorist threat on utility tunnels, where the terrorist threat is modeled as the worst-case attack and the attack consequence is measured as the vulnerability of those UISSs partially carried by the utility tunnels. Hence, this paper only addresses the vulnerability problem during the operation phase of the utility tunnels and the UISSs, and the disruptive event of concern is not arising from one of the systems but is external of those underground systems. Furthermore, this paper will study how the worst-case vulnerability could be mitigated by protecting critical utility tunnel segments and infrastructure components. To the authors' knowledge, none of existing studies on interdependent infrastructure systems have captured the geographical interdependencies due to utility tunnels. Addressing this problem requires particular network-based description model for coupled utility tunnels and UISSs, which will be presented in the next section. However, in the literature, some researchers have studied the worst-case vulnerability for multiple infrastructure systems with other types of interdependencies or dependencies. Ouyang et al. studied the complementary railway and airline systems (logical interdependency) in China, and used a genetic algorithm to identify worst-case failure scenario [32], but this approach cannot guarantee exact solutions. By considering circle shaped localized attacks on physically interdependent systems, Ouyang proposed an enumeration based approach to find the worst-case localized attack [29] and mitigate the worst-case vulnerability [27]. But in this current work, the number of possible worst-case attacks on utility tunnels and UISSs increases exponentially with the number of attacked components, so the enumeration-based method cannot solve the problem. This paper will introduce a column-and-constraint generation based algorithm to solve the proposed model. Wang et al. [44] studied the worst-case vulnerability mitigation of gas-electric systems, but this study did not consider the bi-directional physical interdependency as well as the geographical interdependency, and only modelled the unidirectional physical dependency of power system on gas system. Besides the above worst-case vulnerability studies on interdependent UISSs, vulnerability assessment and mitigation of interdependent UISSs have been extensively studied under other types of hazards, particularly natural hazards [4]. Two relevant projects are the work carried out within the SYNER-G project [16, 36,37] and the work of the NIST-funded Center for Risk-Based Community Resilience Planning [14,19].

The main contributions of this work include two aspects: (1) propose a tri-level model and introduce its solution algorithm to minimize the worst-case vulnerability of utility tunnels, which carry physically, geographically and logically interdependent infrastructure systems (cyber interdependency can be also easily integrated); (2) analyze both negative and positive effects of constructing utility tunnels in urban area. As different infrastructure components carried inside a tunnel segment can be damaged simultaneously, intuitively, constructing utility tunnels only has negative effects. However, through the analysis, it is found that constructing utility tunnels can also benefit system operator or defender. In addition, the proposed worst-case approach can be easily adapted to a robust optimization model for identifying robust mitigation strategy against natural hazards [42]. Moreover, despite this paper does not consider resilience, which is the ability of a system to withstand internal/external stresses and recover from them [11,14,16,17,21,28,35,40], recent studies show that compared with the optimal resilience-based mitigation strategies, vulnerability-based solutions are near-optimal under both worst-case malicious attack and natural hazards [33]. The rest of the paper is organized as follows: Section 2 formulates the problem. Section 3 introduces the solution algorithm. Section 4 shows the results for real utility tunnels that carry interdependent power and water systems in Tianjin Eco-city, China. Section 5 concludes the findings and gives future work.

2. Problem formulation

This section formulates the mathematical model to mitigate the worst-case vulnerability of urban utility tunnels that carry interdependent UISSs. Here, vulnerability under a disruptive event is quantified as the whole UISSs' functionality drop. Formulating the model needs to first know how utility tunnels are deployed and collocated with UISSs in a region of concern. In utility tunnels' practice, there are no strict rules for their layout, only some suggestions have been proposed by experts. Utility tunnels are usually deployed with high priority in particular places [8,23], such as in central business and commercial districts, along roads with heavy traffic, in arterial and collector streets, along high-voltage power transmission corridors, and in areas with frequent land trenching activities. These rules indicate that not all infrastructure components are carried by the utility tunnels, and only some components located in particular places are inside the tunnels. Hence, the layout of the utility tunnels in a region depends on the geographical distribution of infrastructure systems in that region.

This paper denotes the set of UISSs of concern by κ and models each UISS $k \in \kappa$ by a network $G^k(N^k, E^k)$, with all source nodes denoted by N^S , $k \subseteq N^k$ and all demand nodes represented by N^D , $k \subseteq N^k$. For node $n \in N^k$, its real supply and demand are denoted as s_n^k and d_n^k , respectively, with the maximum supply as $\hat{s}_n^k$ and the required demand as $\hat{d}_n^k$. For edge $e \in E^k$, its real flow is f_e^k and its capacity is $\hat{f}_e^k$, with its origin and destination nodes denoted respectively by $o^k(e)$ and $d^k(e)$. Based on the geographical layout of these UISSs, a network of the utility tunnels that carry them is described by $G^T(N^T, E^T)$, consisting of a set of tunnel nodes N^T and a set of tunnel segments E^T . Here, each tunnel segment should carry different infrastructure edges. To illustrate this concept, an interdependent power and water systems is used and shown in Fig. 1. According to the rule that each tunnel segment should carry different infrastructure edges, five nodes T1, T2, T3, T4 and T5 are required in the tunnel network. The tunnel segment T1-T2 carries edges P2-P3 and W1-W2; the segment T2-T3 carries edges P1-P2, P2-P4 and W1-W2; the segment T3-T4 carries edges P1-P2, P2-P4 and W1-W3; the segment T4-T5 carries edges P2-P4 and W1-W3.

Besides describing the infrastructure components carried by each tunnel segment, it also needs to describe the infrastructure components passing through each tunnel node. For example, three power edges P1-P2, P2-P3, P2-P4 and one water edge W1-W2 pass through tunnel node

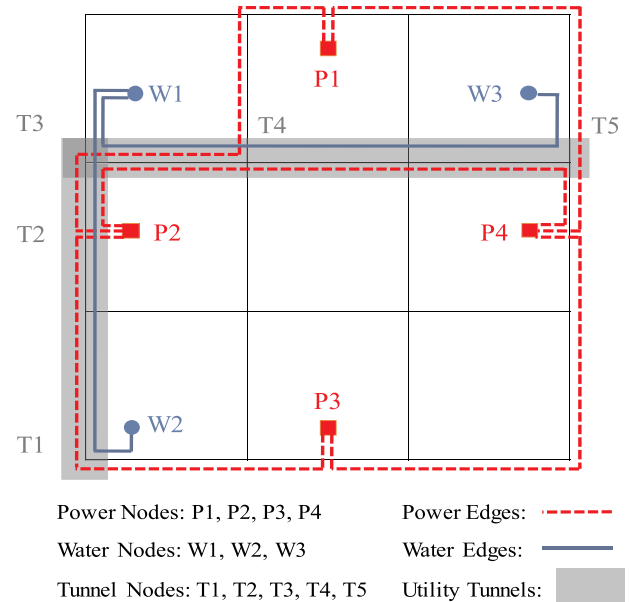


Fig. 1. Geographical layout of a power system, a water system and the utility tunnels in a region.

T2, so when attacking tunnel node T2, it could cause the failure of the four edges. To describe the geographical relationship between the tunnel network and its carried UISs, it defines two matrices $I^k = \{I_{e,n}^k\}$, $J^k = \{J_{e,e'}^k\}$ for each system $k \in \kappa$, where $I_{e,n}^k$ is 1 if edge e in infrastructure k passes through tunnel node n , and 0 otherwise; $J_{e,e'}^k$ is 1 if edge e in infrastructure k is partially carried inside tunnel segment e' , and 0 otherwise. Based on this definition, the matrices I^k and J^k for the two systems in Fig. 1 are written as follows:

$$I^{power} = \begin{matrix} & \begin{matrix} T1 & T2 & T3 & T4 & T5 \end{matrix} \\ \begin{matrix} P1 - P2 \\ P1 - P4 \\ P2 - P3 \\ P2 - P4 \\ P3 - P4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix} J^{power} \quad (1)$$

$$= \begin{matrix} & \begin{matrix} T1 - T2 & T2 - T3 & T3 - T4 & T4 - T5 \end{matrix} \\ \begin{matrix} P1 - P2 \\ P1 - P4 \\ P2 - P3 \\ P2 - P4 \\ P3 - P4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$I^{water} = \begin{matrix} & \begin{matrix} T1 & T2 & T3 & T4 & T5 \end{matrix} \\ \begin{matrix} W1 - W2 \\ W1 - W3 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix} J^{water} \quad (2)$$

$$= \begin{matrix} & \begin{matrix} T1 - T2 & T2 - T3 & T3 - T4 & T4 - T5 \end{matrix} \\ \begin{matrix} W1 - W2 \\ W1 - W3 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

In addition, in Fig. 1, it can be found that some parts of infrastructure edges are carried inside the utility tunnels, but the remaining parts are not carried. If those edges carried inside the utility tunnels are well protected, but those un-carried parts are not protected, this kind of protection is useless as those edges can still be damaged and has the same level of risk compared with the case without protection. To address this feature, it defines a vector $M^k = \{M_e^k\}$ for each system, where M_e^k is 1 if edge e in infrastructure k is not fully carried by utility tunnels, and 0 otherwise. The model for worst-case vulnerability mitigation (WCVM) of urban utility tunnels that carry interdependent UISs is formulated as follows:

Formulation of WCVM Model:

$$\min_{w_n^k, w_e^{k,M}, w_n^T, w_e^T, x_n^k, x_e^k, x_n^T, x_e^T, d_n^k, d_e^k, \delta_{ij}^{k \rightarrow l}} \max_{s_n^k, s_e^k, s_n^T, s_e^T} \min_{\delta_{ij}^{k \rightarrow l}} V = \sum_{k \in \kappa} \eta^k \sum_{n \in N^{D,k}} \left(\hat{d}_n^k - d_n^k \right) \quad (3)$$

Subject to:

$$\sum_{k \in \kappa} \sum_{n \in \tilde{N}^k} c_n^{D,k} w_n^k + \sum_{k \in \kappa} \sum_{e \in \tilde{E}^k \cap M^k} c_e^{D,k,M} w_e^{k,M} + \sum_{n \in \tilde{N}^T} c_n^{D,T} w_n^T + \sum_{e \in \tilde{E}^T} c_e^{D,T} w_e^T \leq B_D \quad (4)$$

$$w_n^T, w_e^T \in \{0, 1\}, \forall n \in \tilde{N}^T, \forall e \in \tilde{E}^T \quad (5)$$

$$w_n^k, w_e^{k,M} \in \{0, 1\}, \forall n \in \tilde{N}^k, \forall e \in \tilde{E}^k \cap M^k, \forall k \in \kappa \quad (6)$$

$$\sum_{k \in \kappa} \sum_{e \in \tilde{N}^k} c_n^{A,k} (1 - x_n^k) + \sum_{k \in \kappa} \sum_{e \in \tilde{E}^k \cap M^k} c_e^{A,k,M} (1 - x_e^{k,M}) + \sum_{n \in \tilde{N}^T} c_n^{A,T} (1 - x_n^T) + \sum_{e \in \tilde{E}^T} c_e^{A,T} (1 - x_e^T) \leq B_A \quad (7)$$

$$x_e^k = \prod_{n \in \{n | I_{e,n}^k = 1\}} [w_n^T + x_n^T (1 - w_n^T)] \prod_{e' \in \left\{ e' | J_{e,e'}^k = 1 \right\}} [w_{e'}^T + x_{e'}^T (1 - w_{e'}^T)] \prod_{e \in \tilde{E}^k \cap M^k} [w_e^{k,M} + x_e^{k,M} (1 - w_e^{k,M})], \forall e \in E^k, \forall k \quad (8)$$

$$x_n^k \geq w_n^k, \forall n \in \tilde{N}^k, \forall k \in \kappa \quad (9)$$

$$x_n^k, x_e^k \in \{0, 1\}, \forall n \in \tilde{N}^k, \forall e \in \tilde{E}^k, \forall k \in \kappa \quad (10)$$

$$x_e^{k,M} \in \{0, 1\}, \forall e \in \tilde{E}^k \cap M^k, \forall k \in \kappa \quad (11)$$

$$x_n^T, x_e^T \in \{0, 1\}, \forall n \in \tilde{N}^T, \forall e \in \tilde{E}^T \quad (12)$$

$$s_n^k + \sum_{e \in \{e | d^k(e) = n\}} f_e^k - \sum_{e \in \{e | o^k(e) = n\}} f_e^k = d_n^k, \forall n \in N^k \quad (13)$$

$$-f_e^k x_e^k x_{o^k(e)}^k x_{d^k(e)}^k \leq f_e^k \leq x_e^k x_{o^k(e)}^k x_{d^k(e)}^k f_e^k, \forall e \in E^k, \forall k \quad (14)$$

$$0 \leq s_n^k \leq x_n^k \hat{s}_n^k, \forall n \in N^{S,k}, \forall k \in \kappa \quad (15)$$

$$0 \leq d_n^k \leq x_n^k \hat{d}_n^k, \forall n \in N^{D,k}, \forall k \in \kappa \quad (16)$$

$$d_i^k = \delta_{ij}^{k \rightarrow l} \hat{d}_i^k, \forall (i, j) \in F_{ij}^{k \rightarrow l} \quad (17)$$

$$s_j^k \leq \delta_{ij}^{l \rightarrow k} \hat{s}_j^k, \forall (i, j) \in F_{ij}^{l \rightarrow k}, j \in N^{S,k} \quad (18)$$

$$d_j^k \leq \delta_{ij}^{l \rightarrow k} \hat{d}_j^k, \forall (i, j) \in F_{ij}^{l \rightarrow k}, j \in N^{D,k} \quad (19)$$

$$f_e^k \leq \delta_{ij}^{l \rightarrow k} \hat{f}_e^k, \forall (i, j) \in F_{ij}^{l \rightarrow k}, \forall e \in E_j^{l \rightarrow k} \cup E_j^{O,k} \quad (20)$$

$$\delta_{ij}^{k \rightarrow l} \leq x_{ij}^l, \forall (i, j) \in F_{ij}^{k \rightarrow l} \quad (21)$$

As the problem in this paper needs to model the operation of each UIS and the physical interdependencies among UISs, which have been addressed in the authors' previous work [27], so those assumptions related to these two parts will be briefly introduced.

Assumption 1. Similar with many existing studies [27,46], this paper simply defines the functionality level of each UIS as the total satisfied demand, then system vulnerability under a disruptive event is expressed as $\sum_{n \in N^{D,k}} (\hat{d}_n^k - d_n^k)$. As UISs have different units for their demand and have varied importance levels concerned by stakeholders, a vulnerability weighting factor η^k is introduced for each UIS to unify the vulnerability units (through adding $\sum_{n \in N^{D,k}} \hat{d}_n^k$ in the denominator) and account for varied importance of different UISs. For example, if two UISs $k = 1, 2$ are equally important, η^k can be set as $\eta^1 = 0.5 / \sum_{n \in N^{D,1}} \hat{d}_n^1$, $\eta^2 = 0.5 / \sum_{n \in N^{D,2}} \hat{d}_n^2$. Note that the total vulnerability of interdependent UISs is not necessarily additive, but this is outside the scope of this work and can be an interesting direction for future research. In addition, the proposed approach is also valid for other objective functions, such as the number of affected customers, and the operational cost, which can be also used as a criterion to guide the defender mitigating and responding to the attack.

Assumption 2. The proposed model assumes that (a) the protection strategy is decided and implemented before the attack with the goal of minimizing the worst-case vulnerability; (b) the attack occurs after the protection with the goal of maximizing the vulnerability, and attacking those protected components does not cause their failure; (c) after the attack, system operator or defender can optimize component flow to minimize the vulnerability. These assumptions together lead to a tri-level min-max-min model. Solving this model can return the optimum mitigation strategy against the worst-case failure, and also the worst-case failure scenario under the pre-event optimum mitigation.

For the defender, this paper only considers one extensively adopted ex-ante defense strategy [3,6], which is to protect critical or vulnerable components so that they are invulnerable to be damaged. This assumption of perfect protection makes sense in some disruption scenarios, like terrorist attacks, as well-protected components can make them almost impossible to be attacked. However, under some disruption scenarios, such as natural hazards, protecting components only reduces their damage probabilities. By sampling or identifying a limited but sufficient set of damage scenarios before and after the protection,

the defend decision can be modeled by constraints similar to perfect protection but with different constraints for system operation, as presented by Ouyang et al. [33] in a recent work. For the problem considered in this paper, there are mainly four types of components that can be damaged by the attacker and then are regarded as candidate components for protection. The first two types of components are those damageable nodes $\tilde{N}^k \subseteq N^k$ in each system and those damageable edges that are not fully carried by utility tunnels, denoted as $M^k \cap \tilde{E}^k$. The second two types of components are those damageable tunnel nodes $\tilde{N}^T \subseteq N^T$ and damageable tunnel segments $\tilde{E}^T \subseteq E^T$. For these four types of components, define binary variables $w_n^k, w_e^{k,M}, w_n^T, w_e^T$ to indicate whether the defender protects the damageable components in each type, respectively. The costs for protecting node n in infrastructure k , protecting not-fully-carried edge e in infrastructure k and protecting tunnel node n and tunnel segment e are respectively denoted by $c_n^{D,k}, c_e^{D,k,M}, c_n^{D,T}$ and $c_e^{D,T}$. The defense budget is denoted by B_D . Then, the defense decision can be described by Constraints (4)–(6). Constraint (4) limits the defense investment. Constraints (5) and (6) enforce binary defense. Note that when the defend cost for each component is identical and set as one, the defense budget is just the maximum number of components to be protected.

For the attacker, define binary variable x_n^k that is 0 if damageable node n in infrastructure k is attacked, and 1 otherwise; binary variable $x_e^{k,M}$ that is 0 if damageable edge e that is not fully carried by utility tunnels in infrastructure k is attacked, and 1 otherwise; binary variable x_n^T that is 0 if damageable tunnel node n is attacked, and 1 otherwise; binary variable x_e^T that is 0 if damageable tunnel segment e is attacked. As attacking different damageable components may require varied efforts from the attacker, to make the attacker's model general, the proposed model assumes varied costs to attack different components, and defines the total attack resources or efforts that the attacker has as the attack budget B_A . Then, the attack decision can be described by Constraint (7), where the costs of attacking node n in infrastructure k , edge e not fully carried by utility tunnels in infrastructure k , node n and edge e in the tunnel network, are denoted by $c_n^{A,k}, c_e^{A,k,M}, c_n^{A,T}, c_e^{A,T}$, respectively. If all these attack costs are identical and set as one, the attack budget is just the maximum number of components to be attacked. Note that this worst-case approach can be easily adapted to a robust optimization model for identifying robust mitigation strategy against natural hazards, where components have different damage probabilities or fragilities, depending on the hazard intensities at component sites [42,45]. In addition, for those infrastructure edges passing through a tunnel segment or node, if this tunnel segment or node is attacked, then all related infrastructure edges are damaged. Hence, for an infrastructure edge, its failure state depends on the damage state of each tunnel segment that carries a part of this edge, the damage state of each tunnel node that this edge passes through and also the damage state of the part of this edge that is not carried by the tunnels. The relationship between the damage state of an edge and the damage states of all its related components is described by Constraint (8). For a protected infrastructure node, it cannot be damaged, as described by Constraint (9). Constraints (10)–(12) enforce binary states of damageable components.

Assumption 3. For each UIS, this paper uses the network flow model for system operation [18, 26]. In this model, the decision variables in infrastructure k include the real node supply s_n^k , the real node demand d_n^k and the real edge flow f_e^k . Their constraints are described by Constraints (13)–(16). Constraint (13) ensures flow conservation at each node. Constraint (14) states the flow limit upon each edge. Constraint (15) defines the largest supply for each source node and Constraint (16) states the required demand for each load node.

For the physical interdependencies among UISs, they are described by a group of ordered component pairs $(i, j) \in F_{ij}^{k \rightarrow l}$, which means the dependencies of component j in system l on node i in another system k . Component j will fail if node i cannot provide sufficient supply. Note that physical interdependencies between two systems (not between two

components) do not mean an ordered component pair (i, j) has bi-directional dependencies from each other, but indicates that there exist some components in one system physically depending on components in the other system, while there are some components in the latter system physically depending on components in the former system. As the physical dependency of an edge on a node can be also modeled as a node-to-node dependency through adding a new node in the edge, without loss of generality, this paper only considers the physical dependency for a node-to-node pair. Define a binary variable $\delta_{ij}^{k \rightarrow l}$, which equals to 1 if the physical dependency of node j in infrastructure l on node i in infrastructure k can operate normally, and 0 otherwise. Then, the physical interdependencies among different UISs are described by Constraints (17)–(20). For each node pair $(i, j) \in F_{ij}^{k \rightarrow l}$, the demand of node i should be either zero or full level, depending on whether their dependency relationship can work normally, as stated by Constraint (17). Also, for each node j in node pair $(i, j) \in F_{ij}^{k \rightarrow l}$, depending on its node type, its upper bound constraint is different. If it is a source node, the constraint is (18); if it is a demand node, the constraint is (19). If node j is damaged, all its attached inflow edges $E_j^{I,k}$ and outflow edges $E_j^{O,k}$ will fail and their real flow is zero, as described by Constraint (20).

For the geographical interdependency, this paper considers simultaneous damage of all infrastructure components passing through an unprotected and attacked tunnel segment or node, which is described by Constraint (8). This paper considers the following logical interdependency among UISs. For each node pair $(i, j) \in F_{ij}^{k \rightarrow l}$ with physical dependency, if this physical dependency does not work, the dependent source node i has zero demand, as described by Constraint (17). This simplified modeling is based on the assumption that source node i only provides services to component j . If source node i has other customers and only a part of its demand is used by component j , it can add a new source node i' in system k , which connects with node i and has the demand equal to what component j requires, then constraint (17) is still valid. This technique is used later in the case study. Note that this model does not consider the backup system [1], which can provide services to component j after the interruption of (i, j) . But this can be integrated into the model by modeling the backup system also as an infrastructure system. In addition, if the dependent demand node j is damaged, the physical dependency $(i, j) \in F_{ij}^{k \rightarrow l}$ does not work, then $\delta_{ij}^{k \rightarrow l}$ becomes zero, as described by Constraint (21). In addition, the cyber interdependency is not considered here, but it can be modelled similarly to the physical interdependency.

3. Solution algorithm

If we denote system operational variables s_n^k, d_n^k, f_e^k together by a vector $\mathbf{y}$, the terrorists' attack variables $x_n^k, x_e^k, x_e^{k,M}, x_n^T, x_e^T$ together by a vector $\mathbf{x}$, and the system defense variables $w_n^k, w_e^{k,M}, w_n^T, w_e^T$ together by a vector $\mathbf{w}$, the model in Section 2 is $\min_{\mathbf{w}} \max_{\mathbf{x}} \min_{\mathbf{y}} \delta V(\mathbf{w}, \mathbf{x}, \mathbf{y}, \delta)$. To solve this model, this paper uses a Column-and-Constrained generation based decomposition algorithm. This type of algorithm has been extensively used in the literature to solve similar tri-level models [15, 28, 44, 46]. The flow diagram to solve the proposed WCVM model is shown in Fig. 2. It first solves an outer-layer sub-problem $\max_{\mathbf{x}} \min_{\mathbf{y}, \delta} V(\hat{\mathbf{w}}, \mathbf{x}, \mathbf{y}, \delta)$ with a fixed defense $\hat{\mathbf{w}}$, which produces the worst-case attack scenario as well as an upper bound of the worst-case vulnerability $V^* \leq V_{Outer, SP}^*$. The algorithm next solves an out-layer master problem $\min_{\mathbf{w}} \min_{\{\mathbf{y}(t), \{\delta(t)\}\}} V(\mathbf{w}, \{\mathbf{y}(t)\}, \{\delta(t)\})_{t=1,2,\dots,T}$, which produces the optimal defense $\hat{\mathbf{w}}$ under fixed attack strategies $\{\hat{\mathbf{x}}(t)\}_{t=1,2,\dots,T}$ returned through solving the outer-layer sub-problem, and also gives a lower bound of the worst-case vulnerability $V_{Outer, MP}^* \leq V^*$. By repeatedly solving the out-layer sub-problem and the out-layer master problem, the number of attack strategies $\{\hat{\mathbf{x}}(t)\}_{t=1,2,\dots,T}$ increases and the difference between the upper bound of V^* and the lower bound of V^* will reduce to the preset optimality gap.

The outer-layer master problem can be reformulated as a single level minimization model [27], with linear objective function and

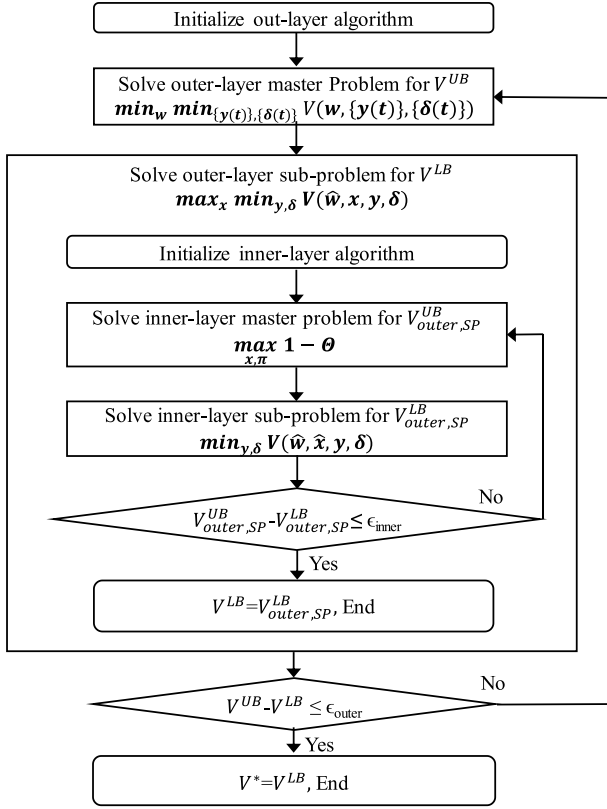


Fig. 2. Flow diagram to solve the proposed WCVM model.

constraints except the non-linear constraint (25) with binary product terms. By using the McCormick method [28], the product terms can be easily linearized. Then, the outer-layer master problem is equivalent to a mixed integer linear programming (MILP) model, which can be directly solved by CPLEX. For the out-layer sub-problem $\max_x \min_{y,\delta} V(\hat{w}, x, y, \delta)$, if the inner layer is a linear programming (LP) model, several algorithms, such as Benders decomposition, can be used for solving it. However, it has binary variable δ , which prevents directly dualizing the inner level problem. Here, a Column-and-Constraint generation based algorithm is used [46]. This algorithm solves the out-layer sub-problem by first selecting an arbitrary attack strategy $\hat{x}$, and then solve an inner-layer sub-problem $\min_{y,\delta} V(\hat{w}, \hat{x}, y, \delta)$. Based on the interdependency operation states $\{\hat{\delta}(h)\}$ solved from the inner-layer sub-problem in each step $h = 1, 2, \dots, H$, the column-and-constrained generation based algorithm solves an inner-layer master problem. This problem is based on the idea that given any interdependency operation states $\hat{\delta}$, the outer-layer sub-problem becomes $\max_x \min_y V(\hat{w}, x, y, \hat{\delta})$. In this max-min formulation, its lower-level becomes a LP model, then it can be rewritten as $\max_x \max_{\pi} V^{Dual}(\hat{w}, x, \pi, \hat{\delta})$, where π is the dual variable of the lower-level LP problem. Based on this idea, given a set of interdependency operation states $\{\hat{\delta}(h)\}$, $h = 1, 2, \dots, H$, the inner-layer master problem is formulated as:

$$\max_{x_n^k, x_e^k, x_c^k, M, x_n^T, x_e^T, \alpha_n^k(h), \beta_e^k(h), \gamma_e^k(h), \mu_n^k(h), \pi_n^k(h), \zeta_n^k(h), \eta_n^k(h), \varphi_n^k(h)} 1 - \Theta \quad (22)$$

Subject to:

$$\begin{aligned} \Theta \geq & \sum_{k \in K} \sum_{e \in E^k} -f_e^k x_e^k x_{\delta(e)}^k \beta_e^k(h) + \sum_{k \in K} \sum_{e \in E^k} \hat{f}_e^k x_e^k x_{\delta(e)}^k \gamma_e^k(h) \\ & + \sum_{k \in K} \sum_{n \in N^{S,k}} \hat{s}_n^k x_n^k \mu_n^k(h) + \sum_{k \in K} \sum_{n \in N^{D,k}} \hat{d}_n^k x_n^k \pi_n^k(h) \\ & + \sum_{k,l \in K} \sum_{(i,j) \in F_{ij}^{l,k}} \hat{d}_{ij}^{D,k} \delta_{ij}^{l,k} s_i^k(h) + \sum_{k,l \in K} \sum_{(i,j) \in F_{ij}^{l,k}} \hat{d}_{ij}^{S,k} \delta_{ij}^{l,k} \eta_j^k(h) \\ & + \sum_{k,l \in K} \sum_{(i,j) \in F_{ij}^{l,k}} \hat{d}_{ij}^{S,k} \delta_{ij}^{l,k} \varphi_j^k(h) \\ & + \sum_{k,l \in K} \sum_{(i,j) \in F_{ij}^{l,k}} \hat{d}_{ij}^{S,k} \delta_{ij}^{l,k} \rho_e^k(h), \forall h \end{aligned} \quad (23)$$

$$\alpha_n^k(h) + \mu_n^k(h) \geq 0, \forall n \in N^{S,k}, C^k, \forall h, \forall k \quad (24)$$

$$\alpha_n^k(h) + \mu_n^k(h) + \eta_n^k(h) \geq 0, \forall n \in N^{S,k} \cap C^k, \forall h, \forall k \quad (25)$$

$$-\alpha_n^k(h) + \pi_n^k(h) \geq \eta^k / \sum_{n \in N^{D,k}} \hat{d}_n^k, \forall n \in N^{D,k} \setminus (C^k \cup D^k), \forall h, \forall k \quad (26)$$

$$-\alpha_n^k(h) + \pi_n^k(h) + \zeta_n^k(h) \geq \eta^k / \sum_{n \in N^{D,k}} \hat{d}_n^k, \forall n \in (N^{D,k} \cap D^k) \setminus C^k, \forall h, \forall k \quad (27)$$

$$-\alpha_n^k(h) + \pi_n^k(h) + \varphi_n^k(h) \geq \eta^k / \sum_{n \in N^{D,k}} \hat{d}_n^k, \forall n \in (N^{D,k} \cap C^k) \setminus D^k, \forall h, \forall k \quad (28)$$

$$-\alpha_n^k(h) + \pi_n^k(h) + \zeta_n^k(h) + \varphi_n^k(h) \geq \eta^k / \sum_{n \in N^{D,k}} \hat{d}_n^k, \forall n \in N^{D,k} \cap C^k \cap D^k, \forall h, \forall k \quad (29)$$

$$\begin{aligned} \alpha_{d(e)}^k(h) - \alpha_{o(e)}^k(h) - \beta_e^k(h) + \gamma_e^k(h) &= 0, \\ \forall e \in E^k \setminus \{e \in E_{ij}^{l,k} \cup E_{ij}^{o,k}, \forall (i,j) \in F_{ij}^{l,k}\}, \forall h, \forall k \end{aligned} \quad (30)$$

$$\begin{aligned} \alpha_{d(e)}^k(h) - \alpha_{o(e)}^k(h) - \beta_e^k(h) + \gamma_e^k(h) + \rho_e^k(h) &= 0, \\ \forall e \in \{e \in E_{ij}^{l,k} \cup E_{ij}^{o,k}, \forall (i,j) \in F_{ij}^{l,k}\}, \forall h, \forall k \end{aligned} \quad (31)$$

$$\begin{aligned} \alpha_n^k(h) \text{ free}, \beta_e^k(h) \leq 0, \gamma_e^k(h) \geq 0, \mu_n^k(h) \geq 0, \pi_n^k(h) \\ \geq 0, \zeta_n^k(h) \text{ free}, \eta_n^k(h) \geq 0, \varphi_n^k(h) \geq 0 \end{aligned} \quad (32)$$

The above model includes dual variables for each constraint in the lower-level outer-layer sub-problem $\max_x \min_y V(\hat{w}, x, y, \hat{\delta})$ given the interdependency operation state variable $\hat{\delta}$, including $\alpha_n^k(h)$ for Constraint (13), $\beta_e^k(h)$ for the left inequality and $\gamma_e^k(h)$ for the right inequality of Constraint (14), $\mu_n^k(h)$ for Constraint (15), $\pi_n^k(h)$ for Constraint (16), $\zeta_n^k(h)$ for Constraint (17), $\eta_n^k(h)$ for Constraint (18), $\varphi_n^k(h)$ for Constraint (19), $\rho_e^k(h)$ for Constraint (20). In addition, C^k is the set of all nodes in infrastructure k , depending on other infrastructure nodes to normally work, and D^k is the set of all nodes in infrastructure k which other infrastructure systems depend on. In the above inner-layer master problem, non-linear bilinear terms in Constraint (34) can be also linearized through applying the McCormick method, with the upper bound of each non-negative or free decision variable as a big positive value $bigM$, and the lower bound of each free variable as $-bigM$. Through this method, the above inner-layer master problem is also equivalent to a MILP and can be directly solved by CPLEX. By iteratively solving the above inner-layer sub-problem and inner-layer master problem, the out-layer sub-problem can then be solved. The procedures to solve the proposed WCVM model are provided as follows:

- (1) Set the initial minimal value of the worst-case vulnerability $V^{LB} \leftarrow 0$, the initial maximal value of the worst-case vulnerability $V^{UB} \leftarrow 1$, the iteration index $T = 1$, and a feasible attack x^T .
- (2) Solve the outer-layer master problem under the fixed attack strategy x^t , $t = 1, \dots, T$, return the optimal objective value $V_{outer,MP}^*$ and the optimal defense w^* , and recalculate the lower bound $V^{LB} \leftarrow V_{outer,MP}^*$.
- (3) Solve the out-layer sub-problem under the fixed defense w^* produced in step 2 according to the following sub-steps, return the

- optimal objective value $V_{outer,SP}^*$ and the optimal attack x^* , and recalculate the upper bound $V_{outer,SP}^{UB} \leftarrow \min\{V_{outer,SP}^{UB}, V_{outer,SP}^*\}$.
- (3.1) Set the initial minimal value $V_{outer,SP}^{LB} \leftarrow -\infty$, the initial maximal value $V_{outer,SP}^{UB} \leftarrow +\infty$, the iteration index $H = 1$, and the initial interdependency operation state $\delta^H \leftarrow 0$.
- (3.2) Solve the inner-layer master problem for δ^h , $h = 1, \dots, H$, and return the optimal objective value $V_{inner,MP}^*$ and the optimal solution x^* , and recalculate $V_{outer,SP}^{UB} \leftarrow V_{inner,MP}^*$.
- (3.3) Solve the inner-layer sub-problem where, return the optimal objective value $V_{outer,SP}^*$ and the optimal interdependency operation state δ^* , and recalculate $V_{outer,SP}^{LB} \leftarrow \max\{V_{outer,SP}^{LB}, V_{inner,SP}^*\}$.
- (3.4) If $V_{outer,SP}^{UB} - V_{outer,SP}^{LB} \leq \epsilon_{inner}$, return the optimal solution $(x^*, V_{outer,SP}^*)$ and end. Otherwise, update $h \leftarrow h + 1$, $\delta^h = \delta^*$, and go back to step 3.2.
- (4) If $V_{outer,SP}^{UB} - V_{outer,SP}^{LB} \leq \epsilon_{outer}$, record the best solution (w^*, x^*, V^*) and end. Otherwise, update $t \leftarrow t + 1$, $x^t = x^*$, and go back to step 2.

4. Case study

This paper takes the utility tunnels that carry interdependent power and water systems in Tianjin-Eco city, China, as an example to illustrate the proposed approach. Fig. 3 shows the geographical layout of the electric power system, the water supply system as well as the utility tunnels in the city. For the power system, there are 7 nodes, two of them are gate substations, receiving power from two generators outside the area. For the water supply system, it has 2 large pump stations and 52 demand nodes. This paper first assumes both two pump stations need electricity to operate normally, and then two more power load nodes P8 and P9 are added near the two pump stations to establish the physical interdependencies, i.e., P8 near W1 and connecting with P7 is added, P9 near W2 and connecting with P4 is added. Other settings of the physical interdependencies will be further discussed in the end of this Section. In the experiments, this paper first assumes all power, water nodes and edges are damageable. The experiments are performed on a

laptop with Intel (R) Core (TM) 2 Due 2.5 GHz CPU and 4GB memory. The proposed algorithm is implemented on MATLAB with CPLEX MATLAB toolbox.

The algorithm sets the vulnerability weighting factor as $0.5 / \sum_{n \in N^{D,k}} \hat{d}_n^k$ for each system $k = 1, 2$. In addition, based on the assumption that similar components should have identical attack cost, this paper initially sets one unit cost to attack each damageable component, i.e., $c_n^{A,k}, c_e^{A,k,M}, c_n^{A,T}, c_e^{A,T} = 1.0$. In this case, the worst-case attack scenario gives the most B_A critical components. The cases with varied attack costs for components will be discussed later in this section. In addition, similar components should have the same defense cost, i.e., $c_n^{D,k}, c_e^{D,k,M}, c_n^{D,T}, c_e^{D,T} = 1.0$, which means that the defense budget is the maximum number of components to be protected. The cases with varied defense costs for components will be also discussed later in this section. The difference gap between the lower and the upper bounds of the worst-case vulnerability is set as $\epsilon_{inner} = \epsilon_{outer} = 0.001$.

This small gap value of ϵ makes the results obtained in all experiments exactly optimum. Based on the above parameter settings, this section will first analyze the computational cost of the solution algorithm, then discuss the worst-case vulnerability under various combinations of defense budget B_D and attack budget B_A . Also, the benefit of constructing utility tunnels will be discussed from both the attacker's and defender's perspectives, and the effects of various settings of physical inter-dependencies will be explored.

First, Table 1 illustrates the results for the computational complexity of the solution algorithm. The columns 1 and 2 give different instances by using different values of B_D and B_A . The columns 3–5 show the average computational time required for solving the outer-layer master problem, the inner-layer master problem and the inner-layer sub-problem, respectively. The sixth column displays the required iterations and the seventh column gives the total computational time for each instance and the worst-case vulnerability V^* under each instance is given in the last column. The table shows greatly different computational time for various instances. Because B_D largely influences the

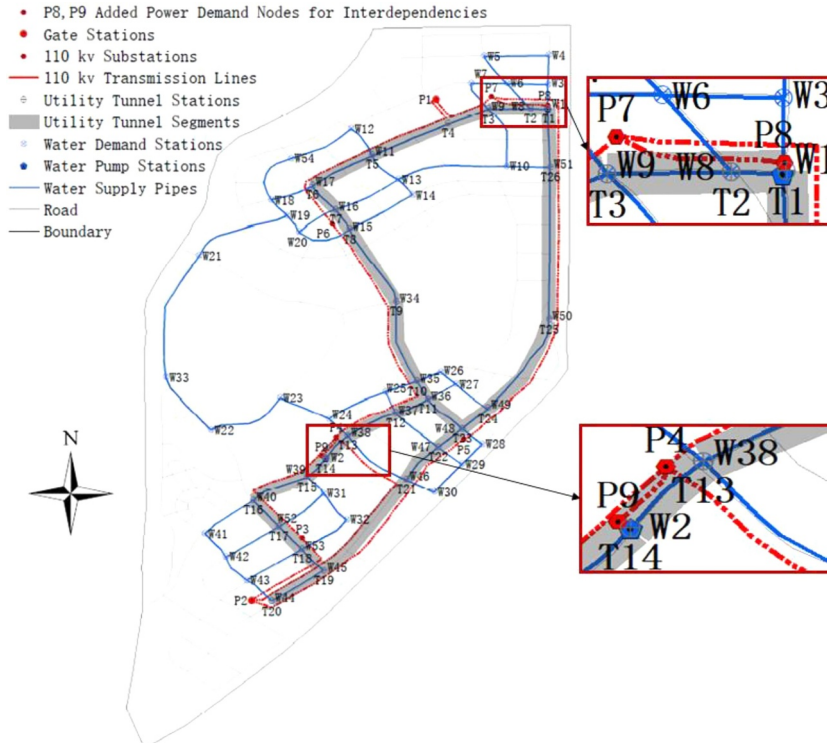


Fig. 3. Geographical Layout of the power, water systems and the utility tunnels in Tianjin Eco-City, China. The physical dependencies of water node W1 on power node P8, and water node W2 on power node P9 are highlighted in the figure.

Table 1
Computational efficiency and computational time to solve the WCV model.

Instances B_D	B_A	Outer-layer master problem Av. time (s)	Inner-layer master problem Av. time (s)	Inner-layer sub-problem Av. time (s)	# of total iter. (Av. Inner iter.)	Total time (s)	V^*
2	1	0.27	1.17	0.05	3 (3.33)	13.18	0.22
	2	0.38	1.45	0.06	4 (4)	25.61	0.55
	3	0.38	0.54	0.05	4(3.33)	9.40	1
4	1	0.41	1.25	0.05	5(3.2)	22.94	0.22
	2	0.96	1.50	0.05	10(3.7)	67.09	0.46
	3	1.13	0.93	0.05	10(3.8)	48.68	0.72
6	1	0.56	1.30	0.05	7(2.7)	29.53	0.22
	2	1.90	1.54	0.05	15(3.53)	112.70	0.41
	3	4.71	1.04	0.06	17(3.7)	149.26	1

solution space of the outer-layer master problem, and B_A affects the solution space of the inner-layer master problem. Note that the exponential growth of the computational time is an intrinsic and common feature of the column-and-constrained generation based solution algorithms [15,28] and the branch-and-cut algorithms. Nonetheless, the proposed worst-case vulnerability mitigation model for urban utility tunnels can be solved off-line as the computational time is not so urgent for those pre-attack mitigation problems.

Second, in the case of no defense, solving the proposed WCV model can return the worst-case attack scenario and the worst-case vulnerability. To analyze the effect of constructing utility tunnels, the interdependent systems without utility tunnels are also considered. Table 2 shows the results. In the table, various costs for infrastructure nodes are considered. From the table, it can be found that when $c_n^{A,k} = 1$, i.e., all nodes can be attacked with the same cost, whether constructing the utility tunnels or not does not affect the worst-case vulnerability, because attacking the two power gate substations can fully collapse the whole system. When $c_n^{A,k} = 2$, there are large differences for the worst-case attack scenarios between the cases with and without utility tunnels. If only attacking one component, utility tunnels increase the worst-case vulnerability by 61.8%, from 0.21 for the case without utility tunnels to 0.34 for the case with utility tunnels. Also, for the case with utility tunnels, attacking two tunnel nodes makes the whole system fully collapsed; but for the case without utility tunnels, it requires attacking four components to fully disrupt the whole system. When $c_n^{A,k} = \infty$, i.e., all infrastructure nodes are invulnerable and only infrastructure edges can be attacked, utility tunnels increase the worst-case vulnerability by at most 81.8% when $B_A = 2$; also, attacking two components can fully collapse the whole system for the case with utility tunnels, but for the case without utility tunnels, it needs to attack at least five components. These results indicate that the negative effect of constructing utility tunnels on the worst-case vulnerability may become more significant if infrastructure nodes are difficult to be attacked or well protected. In the following experiments, this paper assumes all nodes are invulnerable.

Third, when the defense investment is considered, given an attack budget B_A and a defense budget B_D , solving the WCV model can produce the best defense strategy, the minimal worst-case vulnerability and also the worst-case attack scenario under the best defense. Fig. 4(a) shows the minimal worst-case vulnerability curves as a function of B_A under several B_D for the case without utility tunnels; Fig. 4(b) shows similar curves for the case with utility tunnels. From the figures, no matter whether the utility tunnels have been constructed or not, small B_D can significantly reduce the worst-case vulnerability under small attack intensity. For example, protecting one component ($B_D = 1$) reduces the worst-case vulnerability under $B_A = 2$ from 0.55 to 0.39 (mitigation ratio 29.1%) for the case without utility tunnels, and from 1.0 to 0.72 (mitigation ratio 28.0%) for the case with utility tunnels. Protecting three components ($B_D = 3$) reduces the worst-case vulnerability under $B_A = 2$ from 0.55 to 0.25 (mitigation ratio 54.5%) for the

case without utility tunnels, and from 1.0 to 0.55 (mitigation ratio 45.0%) for the case with utility tunnels. When B_A is larger, such as 4 and 5, it shows the same defense budget has significantly less mitigation efficiency (or ratio) for the system with utility tunnels. For example, protecting three components reduces the worst-case vulnerability under $B_A = 4$ from 0.84 to 0.51 for the system without utility tunnels, but zero reduction for the case with utility tunnels. Protecting five components reduces the worst-case vulnerability under $B_A = 5$ from 1.0 to 0.43 for the system without utility tunnels, but only from 1.0 to 0.95 for the system with utility tunnels. These results indicate it is more difficult to protect the system with utility tunnels under large attack intensity. The top figure in Fig. 4(c) shows the difference of the worst-case vulnerability for the systems with and without utility tunnels under different combinations of B_D and B_A . From the figure, it can be found that given any defense budget, it always has a peak for the worst-case vulnerability difference, but occurs at different attack intensity. In the case of no defense, the utility tunnels affect the worst-case vulnerability the most when attacking two components, and in the case of protecting three components, the difference of the worst-case vulnerability can be 0.62 under $B_A = 3$, and in the case of protecting nine components, the largest worst-case vulnerability difference is 0.58 under $B_A = 5$. If analyzing the worst-case vulnerability increase in percentage (or aggravation ratio), the bottom figure in Fig. 4(c) shows the utility tunnels can increase the worst-case vulnerability by 827.74% at $B_A = 2$ and $B_D = 9$ (the aggravation ratio is ∞ at $B_A = 1$ and $B_D = 9$, as the worst-case vulnerability without utility tunnels is 0), while larger defense budget increases the aggravation ratio of the worst-case vulnerability. In sum, constructing utility tunnels may significantly aggravate the

Table 2
Worst-case attack scenarios and their associated worst-case vulnerability in the case of no defense.

With Utility Tunnels						
B_A	$c_n^{A,k} = 1$	V^*	$c_n^{A,k} = 2$	V^*	$c_n^{A,k} = \infty$	V^*
	Worst-case attack		Worst-case attack		Worst-case attack	
1	P1	0.34	T4	0.34	T4	0.34
2	P1, P2	1	T4, T19	1	T4, T19	1
Without Utility Tunnels						
	$c_n^{A,k} = 1$	V^*	$c_n^{A,k} = 2$	V^*	$c_n^{A,k} = \infty$	V^*
	Worst-case attack		Worst-case attack		Worst-case attack	
1	P1	0.34	P7-P8	0.21	P7-P8	0.21
2	P1, P2	1	P4-P9, P7-P8	0.55	P4-P9, P7-P8	0.55
3	–	–	P1-P7, P2-P5, P4-P9	0.72	P1-P7, P2-P5, P4-P9	0.72
4	–	–	P1, P2	1	P1-P6, P1-P7, P4-P9, P7-P8	0.84
5	–	–	–	–	P1-P6, P1-P7, P2-P3, P2-P4, P2-P5	1

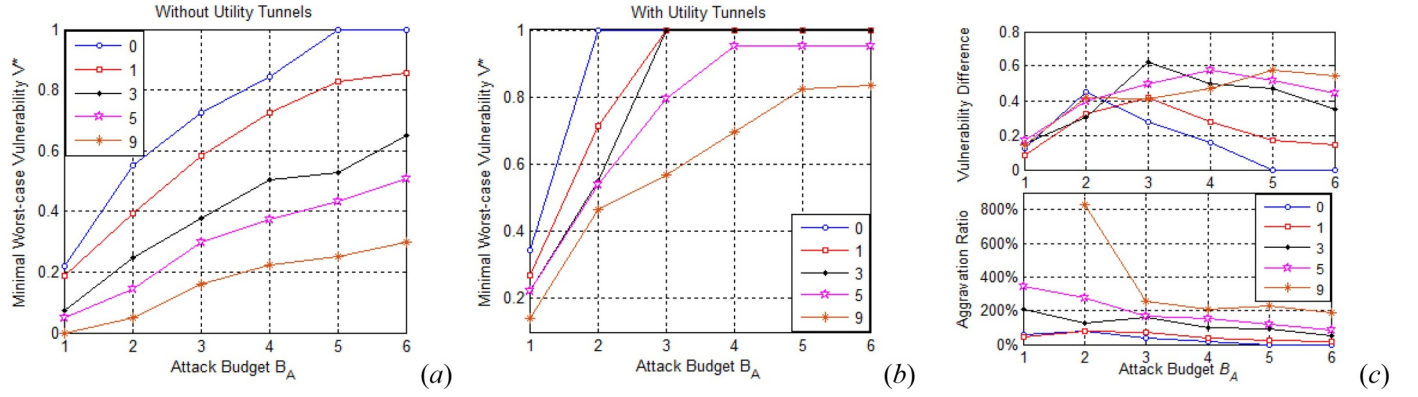


Fig. 4. Minimal worst-case vulnerability as a function of the attack budget under several defense budget B_D for the system without utility tunnels (a) and the system with utility tunnels (b). (c) Top figure: Difference of the minimal worst-case vulnerability as a function of the attack budget under several B_D for the two cases; Bottom figure: aggravation ratio curves of the worst-case vulnerability, computed as the ratio of the difference of the worst-case vulnerability to the worst-case vulnerability without utility tunnels.

worst-case vulnerability, but the aggravation ratio depends on B_A and B_D .

Fourth, the above defense analysis is based on the following parameter settings: each component has the same cost to be protected. In practice, the defense cost for an edge component might largely depend on its length. To consider this, the experiments set the defense cost for each edge component as 1.0 unit per kilometer, while a tunnel node will become invulnerable if any of its attached tunnel segments is defended or protected. This means the defense cost for each tunnel node will be zero, but its defense state is determined according to the following equation:

$$w_n^T = 1 - \prod_{e \in E_n^{I,T} \cup E_n^{O,T}} [1 - w_e^T] \quad (33)$$

This equation can be also linearized by using McCormick method. Based on these parameter settings, solving the modified WCV model can return the best defense strategy, the minimal worst-case vulnerability and the worst-case attack scenario under any combination of B_A and B_D . Fig. 5(a) and Fig. 5(b) shows the worst-case vulnerability curves under several B_D for the two cases, and Fig. 5(c) shows the difference and the aggravation ratio of the worst-case vulnerability curves. From the figures, under small B_D , such as 5, the vulnerability difference curve is always above the x-axis, which means the utility tunnels benefits the attacker. The utility tunnels increase the worst-case vulnerability by at most 88.68% at $B_A = 2$, with an absolute magnitude 0.47 from 0.53 to 1.0. When considering a larger B_D , such as 10, it shows the worst-case vulnerability differences are very small and close to zero, which means the utility tunnels does not or just slightly benefit the attacker. When B_D is large enough, such as 20, the worst-case vulnerability difference

could be negative, which means the utility tunnels does not benefit the attacker but benefit the defender. For example, to mitigate the worst-case vulnerability under $B_A = 6$, investing the same defense budget $B_D = 20$ reduce the worst-case vulnerability 0.1, with a decrease ratio around 50%, compared to the system without utility tunnels.

Fifth, the above studies are based on the two physical dependencies of the power system on the water system. A raised question is how various physical interdependencies between the two systems affect the worst-case vulnerability and the contribution level of the utility tunnels to the worst-case vulnerability. To address this problem, this paper further considers two more cases for the physical interdependencies (case 1 is the current interdependency settings): Case 2: no physical interdependency between the two systems; Case 3: additionally consider two physical interdependencies from water node W7 to power node P1, and water node W44 to power node P2. Fig. 6(a) shows the worst-case vulnerability for the system with utility tunnels as a function of the attack intensity under $B_D = 5$. From the figure, it can be found that with the intensity increase of the physical interdependencies between the two systems, the whole system becomes more and more fragile. For example, under $B_A = 5$, the worst-case vulnerability with utility tunnels is 0.46 in the case of no physical interdependency, and increases to 0.69 in the case of two physical dependencies of water system on power system, and reaches 0.83 in the case of bi-directional physical interdependencies. Similar results can be also found for other B_D , and also for the systems without utility tunnels. Fig. 6(b)–(d) show the difference and aggregation ratio curves of the worst-case vulnerability as a function of the attack budget under three typical B_D . These figures show how the physical interdependencies affect the contribution level of utility tunnels to the worst-case vulnerability. In the case of no

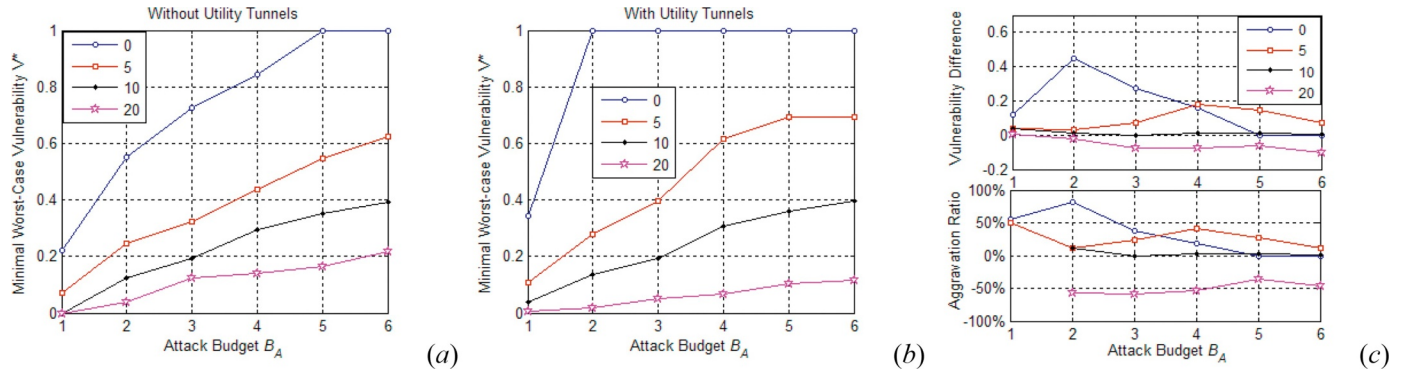


Fig. 5. Minimal Worst-case vulnerability as a function of the attack budget under several defense budget B_D for the system without utility tunnels (a) and the system with utility tunnels (b), when the defense cost for each edge component is proportional to its length. (c) Top figure: Difference of the minimal worst-case vulnerability as a function of B_A under several B_D for the two cases; bottom figure: aggravation ratio curves of the worst-case vulnerability.

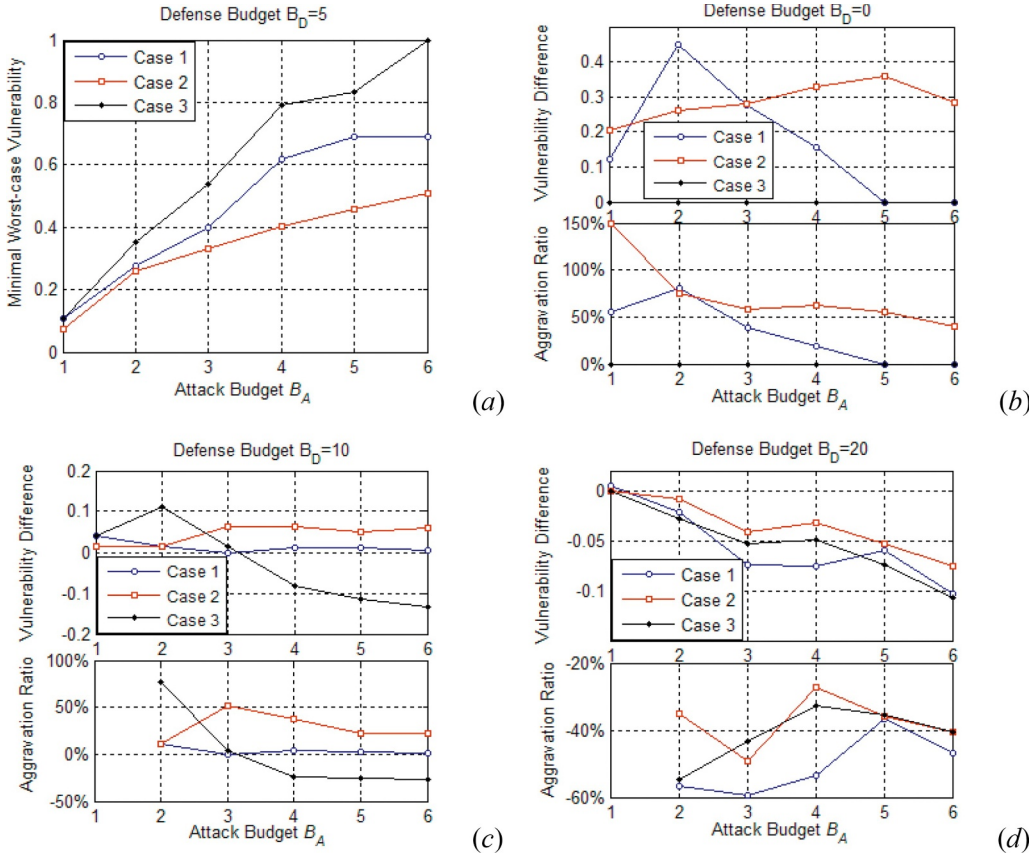


Fig. 6. (a) Minimal worst-case vulnerability as a function of the attack intensity under defense budget $B_D = 5$ for various settings of the physical interdependencies. (b)–(d): Difference and aggregation ratio of the minimal worst-case vulnerability as a function of the attack budget for various settings of the physical interdependencies; (b) $B_D = 0$; (c) $B_D = 10$; (d) $B_D = 20$.

defense $B_D = 0$, from Fig. 6(b), it can be found that the utility tunnels affect the case system 2 the most at $B_A = 5$ from the perspective of the absolute increased value (0.36) of the worst-case vulnerability and at $B_A = 1$ from the perspective of the aggregation ratio (150%), but almost do not affect the case system 3. Also, for the case systems 1 and 2, at no defense or small defense budget, the utility tunnels mainly benefit the attacker, with the worst-case vulnerability increasing up to 0.47; but for the case system 3 and in the case of no defense, no matter considering the utility tunnels or not, this system is too fragile and attacking two components can fully collapse the whole system, so the worst-case vulnerability difference is zero or almost zero for any B_A . At around the threshold value of the defense budget (close to 10), as shown in Fig. 6(c), constructing the utility tunnel only slightly affect the worst-case vulnerability, with the difference in the range $[-0.01, 0.05]$ for the case systems 1, 2; but for the case system 3, the utility tunnels benefit the attacker at some attack intensity, such as $B_A = 1, 2, 3$, but benefit the defender at some other attack intensity, such as $B_A = 4, 5, 6$. Above that threshold defense budget, as shown in Fig. 6(d), the utility tunnels almost always benefit the defender, and with the same defense investment, it is more effective to mitigate the worst-case vulnerability for the systems with utility tunnels. The utility tunnels can reduce the worst-case vulnerability by around 25%–60%.

5. Conclusion and future work

This paper has proposed a mathematical framework to mitigate the worst-case vulnerability of urban utility tunnels that carry interdependent UIs. Real utility tunnels that carry interdependent power and water systems in Tianjin Eco-city, China, have been used to demonstrate the proposed approach. Results show that (1) The utility tunnels affect the worst-case vulnerability more significantly when infrastructure nodes have larger cost to be attacked or they are well protected, this is because the utility tunnels mainly increase the

geographical interdependencies for edge components in different systems; (2) When all infrastructure nodes are invulnerable and the defense costs for all damageable components are identical, the utility tunnels mainly aggravate the worst-case vulnerability, with its absolute value increasing up to 0.62 compared with the system without utility tunnels. (3) When all infrastructure nodes are invulnerable and the defense cost for each edge component is proportional to its length, there exists a threshold defense budget, below which the utility tunnels benefit the attacker and increase the worst-case vulnerability up to 0.47; around which the utility tunnels do not or just slightly affect the worst-case vulnerability; above which the utility tunnels benefit the defender and reduce the worst-case vulnerability up to 0.1. (4) Increasing the intensity level of the physical interdependencies will aggravate the worst-case vulnerability and largely influence the contribution level of the utility tunnels to the worst-case vulnerability. Despite the above results are concluded from small systems, those findings described in a general qualitative form also hold true for large-scale systems. For example, utility tunnels have both negative and positive effects on the worst-case vulnerability of interdependent UIs. If those tunnel segments that carry different system components are not well protected or their exposures are extensive, utility tunnels might increase the worst-case vulnerability; if those tunnel segments have been well protected, utility tunnels might reduce the worst-case system vulnerability. Hence, mitigating the vulnerability of interdependent UIs can be also reached by constructing well-protected utility tunnels.

Numerous issues remain for future work, and they include: (1) This paper only considers the effect of utility tunnels from the perspective of increasing and reducing worst-case vulnerability. In practice, the interdependent UIs might be operated not only by the vulnerability criterion; also, the utility tunnels affect infrastructure repairs [41], and some new technologies and emerging resources can help reduce component fragility [2], so considering these factors and conducting more comprehensive cost-benefit analysis on utility tunnels is an interesting

direction for future research. (2) This paper only considers a fixed utility tunnel layout, then making the tunnel layout also as a decision variable and mitigating the worst-case vulnerability through adjusting the tunnel layout is a direction for future research. (3) This paper only studies the effect of utility tunnels under the worst-case failure with deterministic model parameters, such as the attack and defense costs for different components, considering their uncertainties and performing sampling-based analysis to estimate their effects can be a direction for future research. Also, the worst-case approach can be regarded as a robust optimization to tackle the optimization problems under uncertainty, adapting this approach to natural hazards is also a direction for future research. (4) The proposed approach that ensures the exact solutions needs an exponentially increasing computational time with the problem size. For large-scale systems, if the solutions for the mixed integer linear programming models in the sub-problem and the master problem are beyond the capacity of CPLEX or the computational cost is beyond the acceptable requirement, proposing efficient heuristic methods and checking their solution qualities is also a direction for future research.

Acknowledgments

This material is based upon work supported by China State Grid Corp headquarters science and technology project titled “Research on Regional Integrated Energy Supply Systems Modeling and Planning Techniques”. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the sponsors.

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